

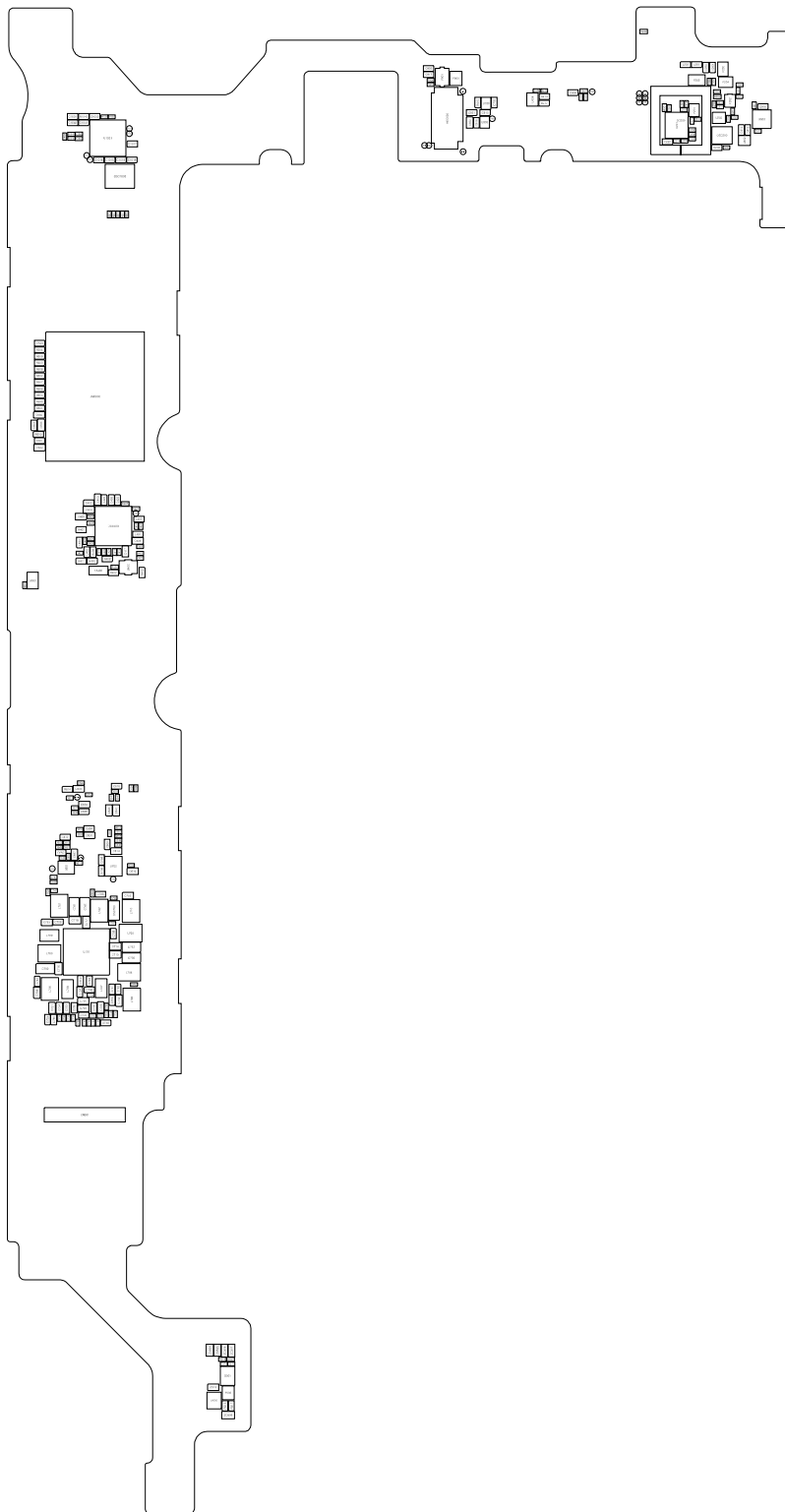
8-1. Block Diagram

The block diagram illustrates the SC54412BHB-A030 AP chip and its various interfaces. Key components and connections include:

- BT/WiFi/GPS:** Includes BT/WiFi (BCM43434HUBG), GPS (BCM4752UB2G), and BT/WiFi/GPS modules. It features antennas (ANT), D-Plexers (D, P, X), and various control and data lines.
- IF:** A 11pin to 44pin IF Connector (MCP145EVCV-1) connecting the AP chip to the PMIC.
- PMIC:** MAX77686 GEWE-Y, responsible for power management. It includes an EXT_IQ_POWER_CONTROL (MAX145EVCV-1) and various power and control lines.
- SENSOR/SM CAM/VT CAM:** Includes 5M CAM CON (P903-215-0.35BW) and 13M VT CAM CON (P903-215-0.35BW). It features a Magnetic Sensor (AK8975) and an ACC Sensor (AK8975).
- AP CHIP:** SC54412BHB-A030, the central processing unit. It includes various internal blocks like BT/WiFi, GPS, IF, PMIC, and AUDIO CODEC.
- AUDIO CODEC:** WM1811 AECS, responsible for audio processing. It includes various audio control and data lines.
- Other Components:** Includes a Connector (P903-215-0.35BW), a 3 SPI EARJACK + SUB MIC (P903-215-0.35BW), and a Left SPK conn. (P903-215-0.35BW).

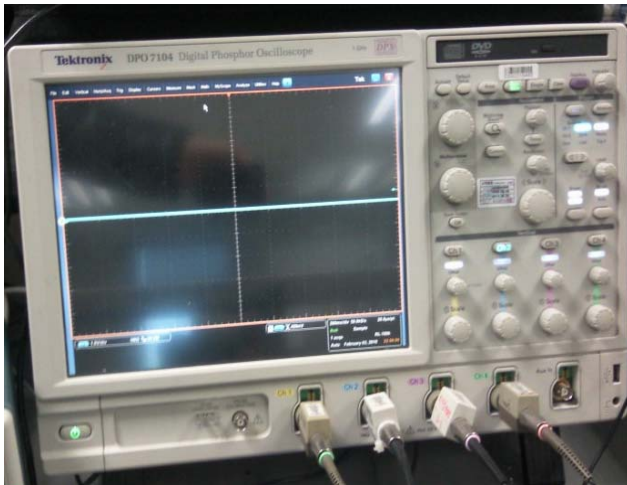
8-2. PCB Diagrams

8-2-1. Top



8-3. Flow Chart of Troubleshooting

Equipments



↑ Oscilloscope



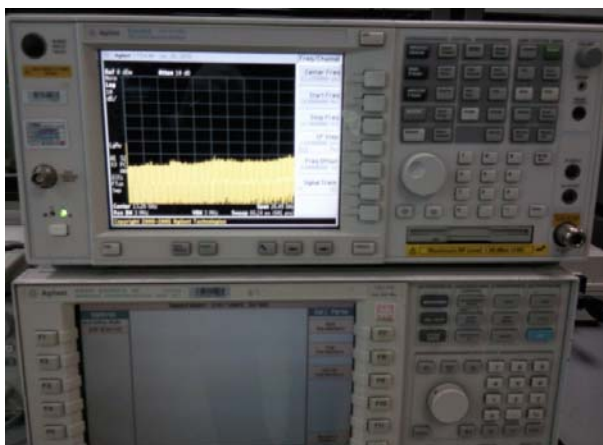
↑ Digital Multimeter



↑ Power Supply



↑ + driver, ESD Safe Tweezer

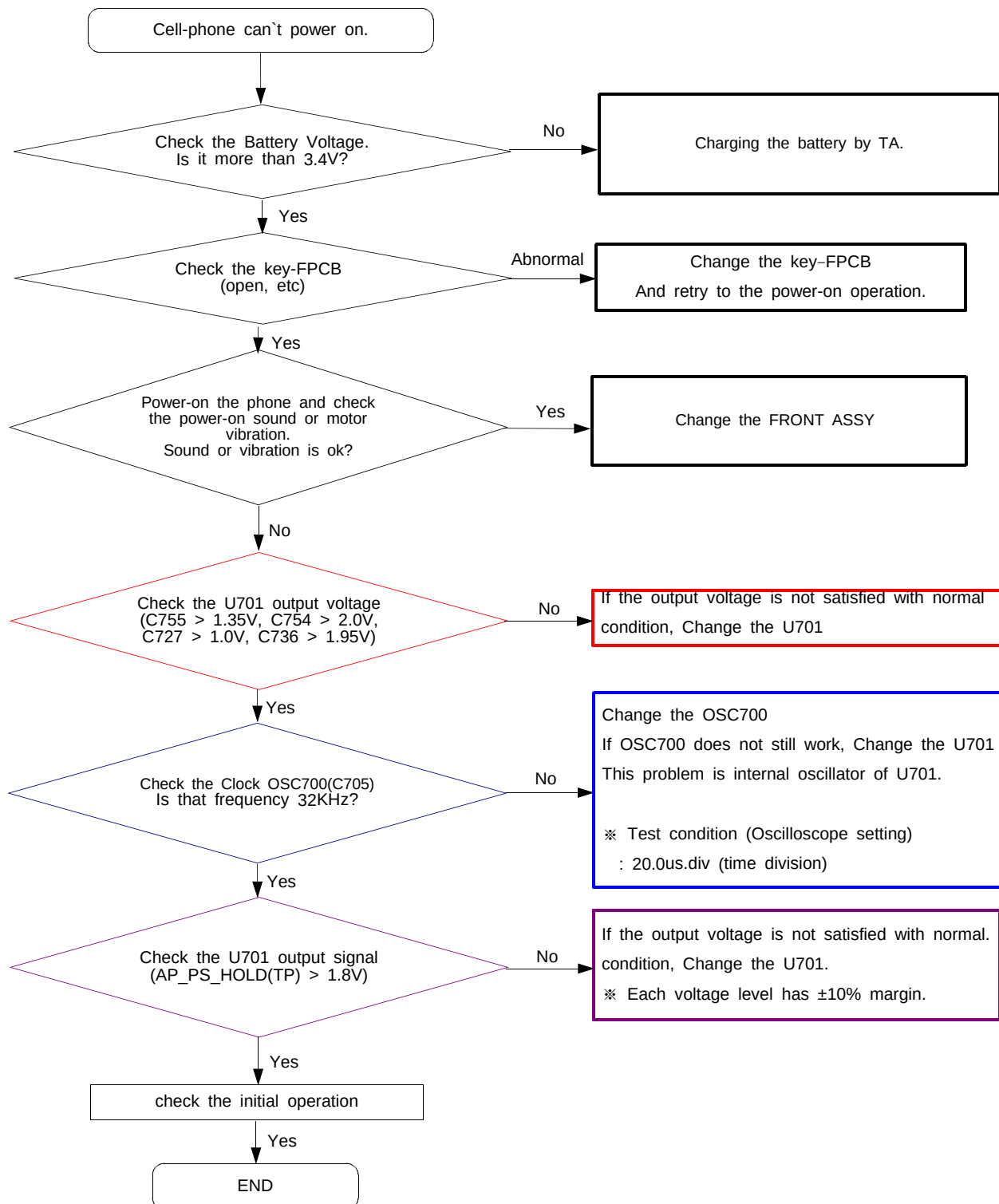


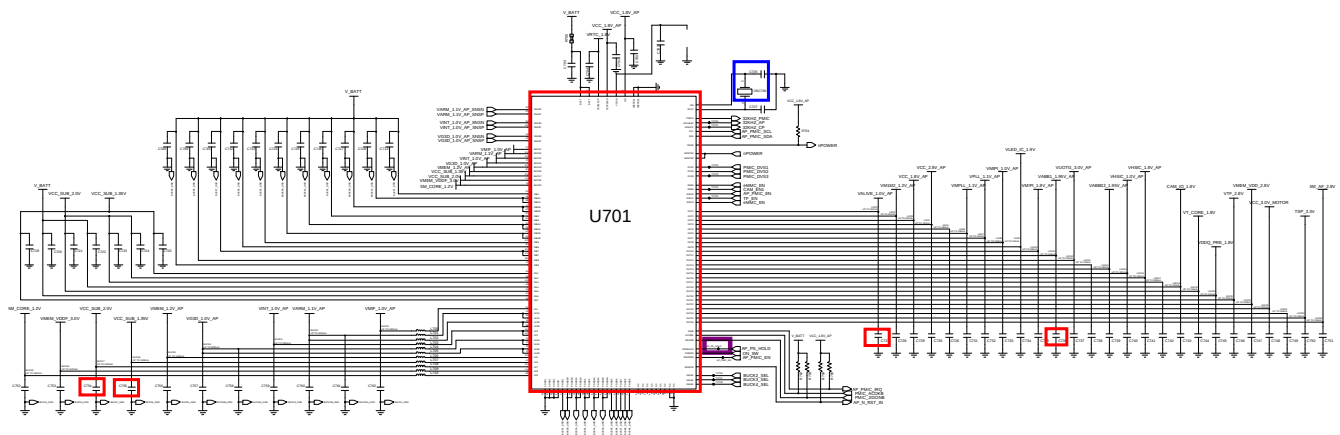
↑ 8960 & Spectrum Analyzer



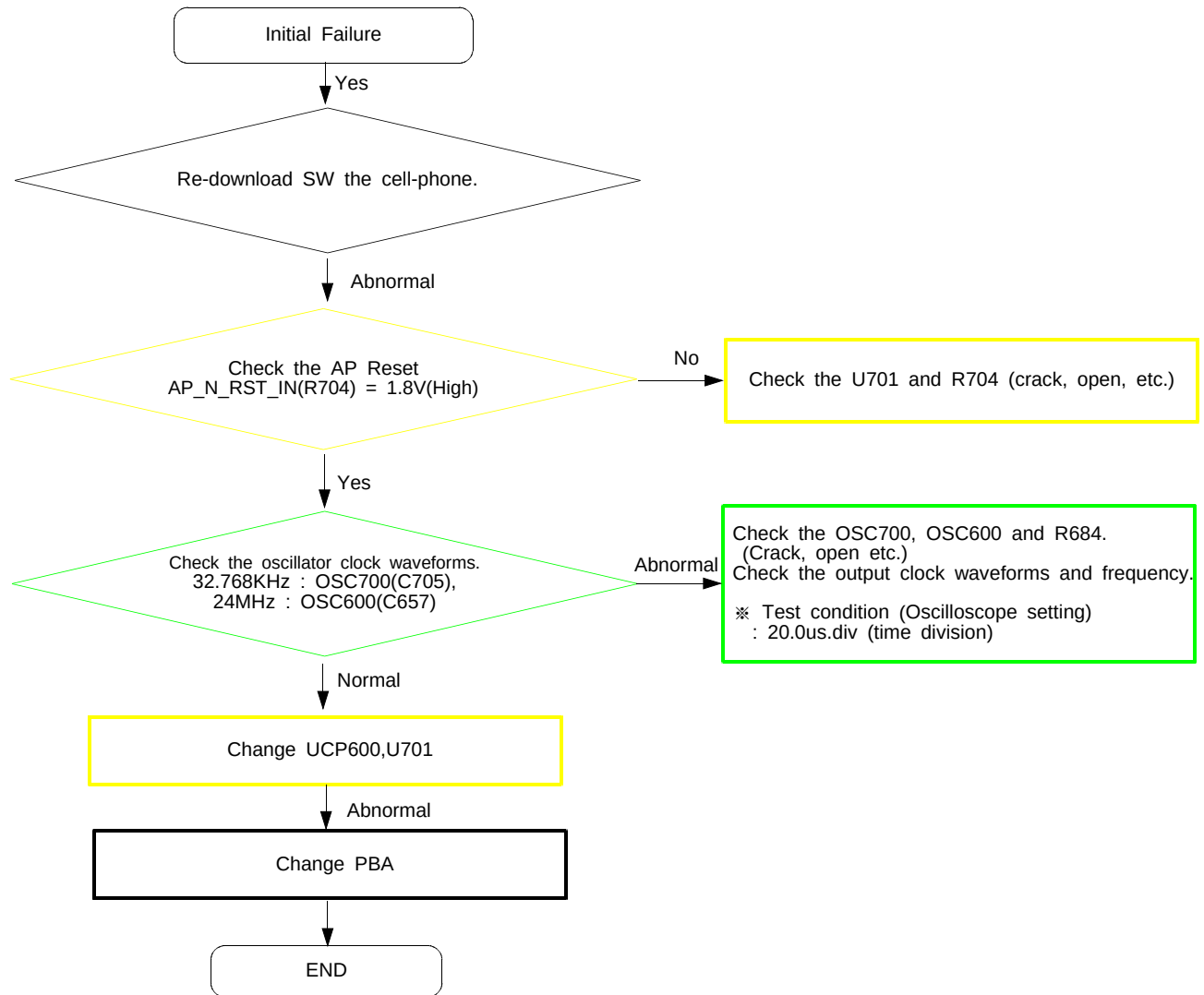
↑ Soldering iron

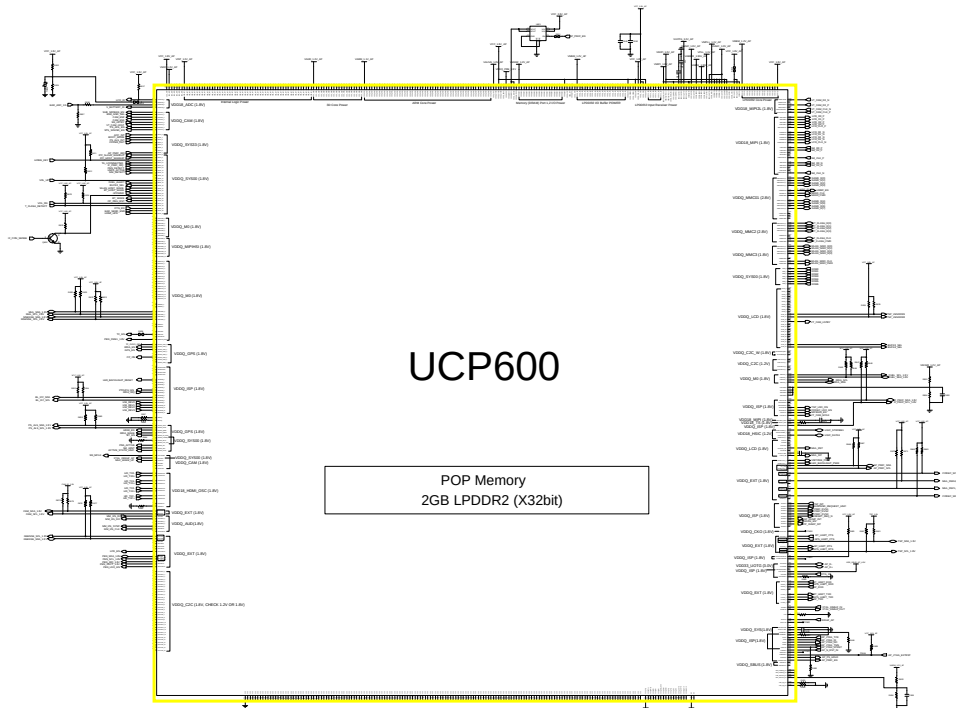
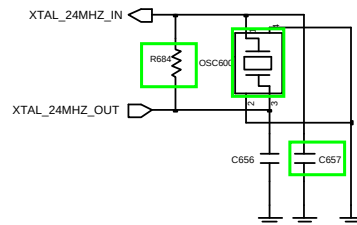
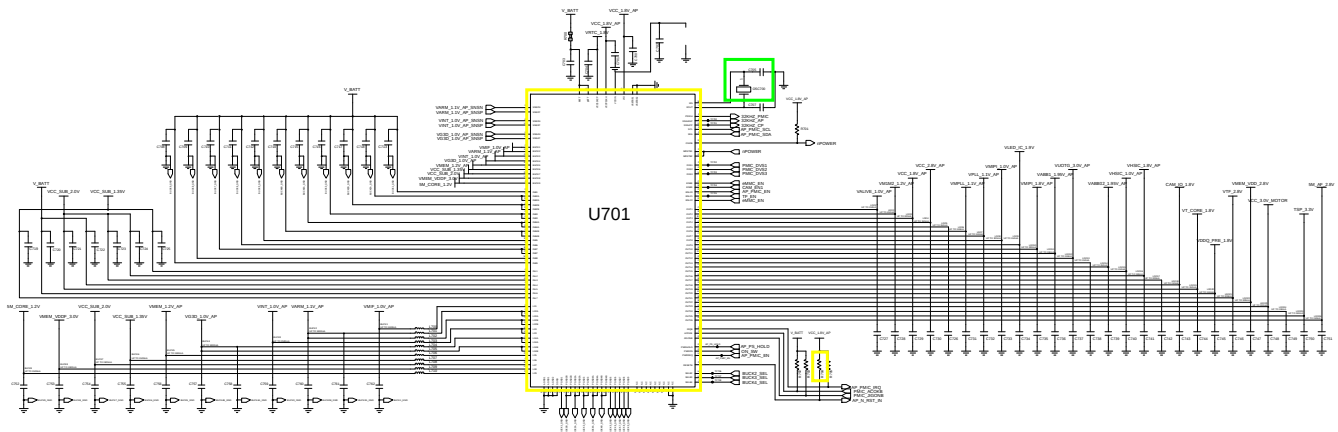
8-3-1. Power On





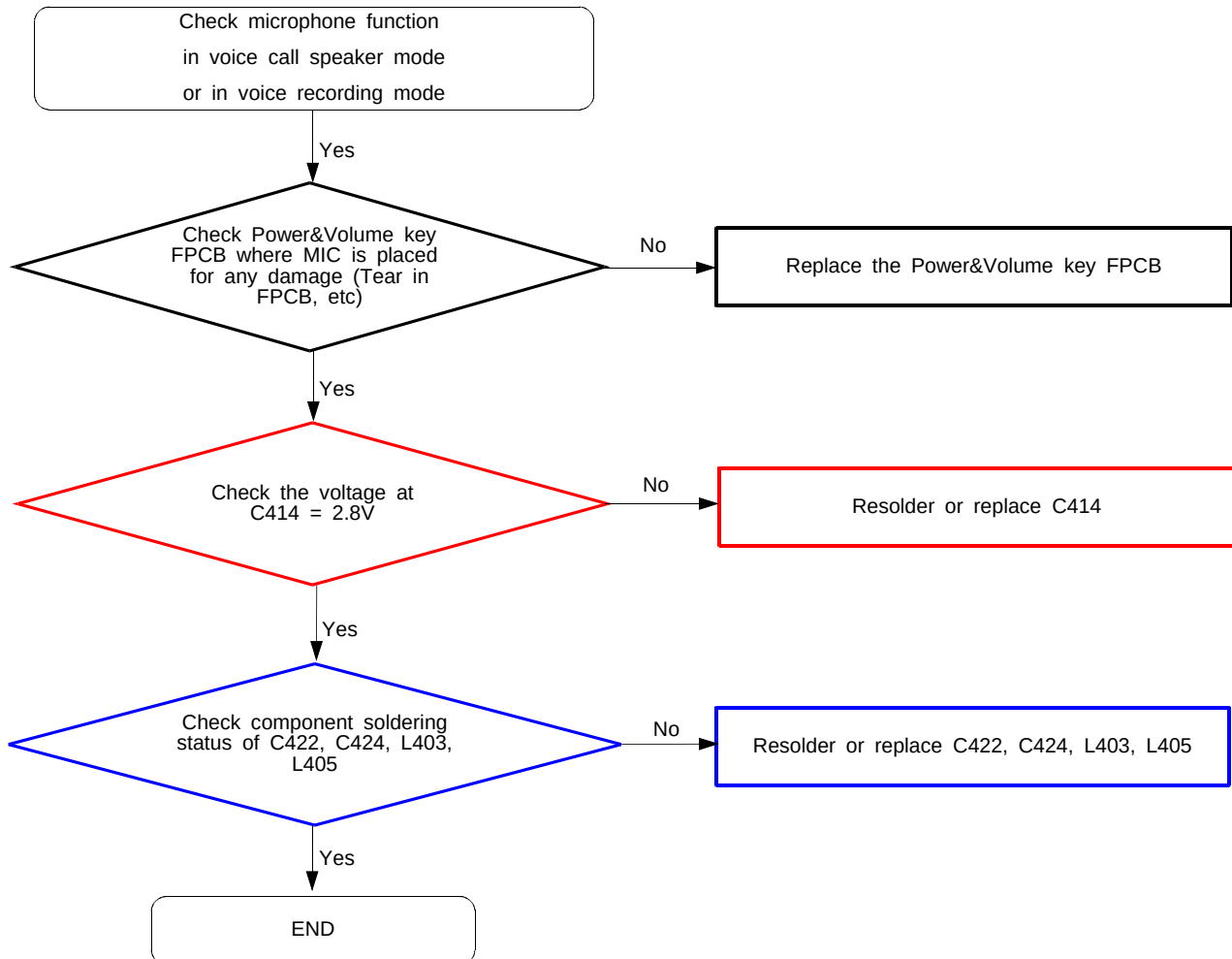
8-3-2. Initial

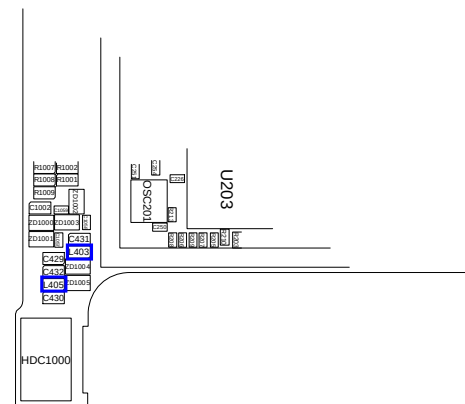
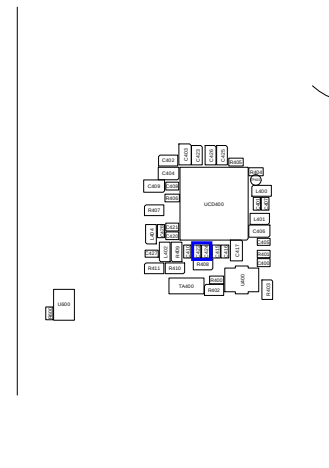
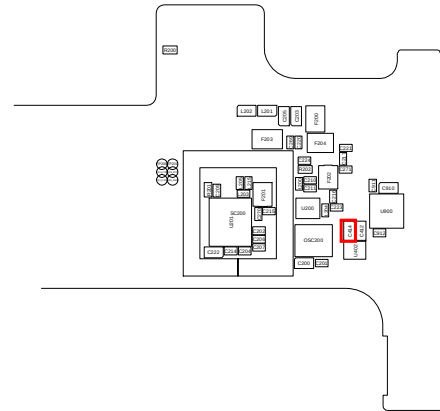




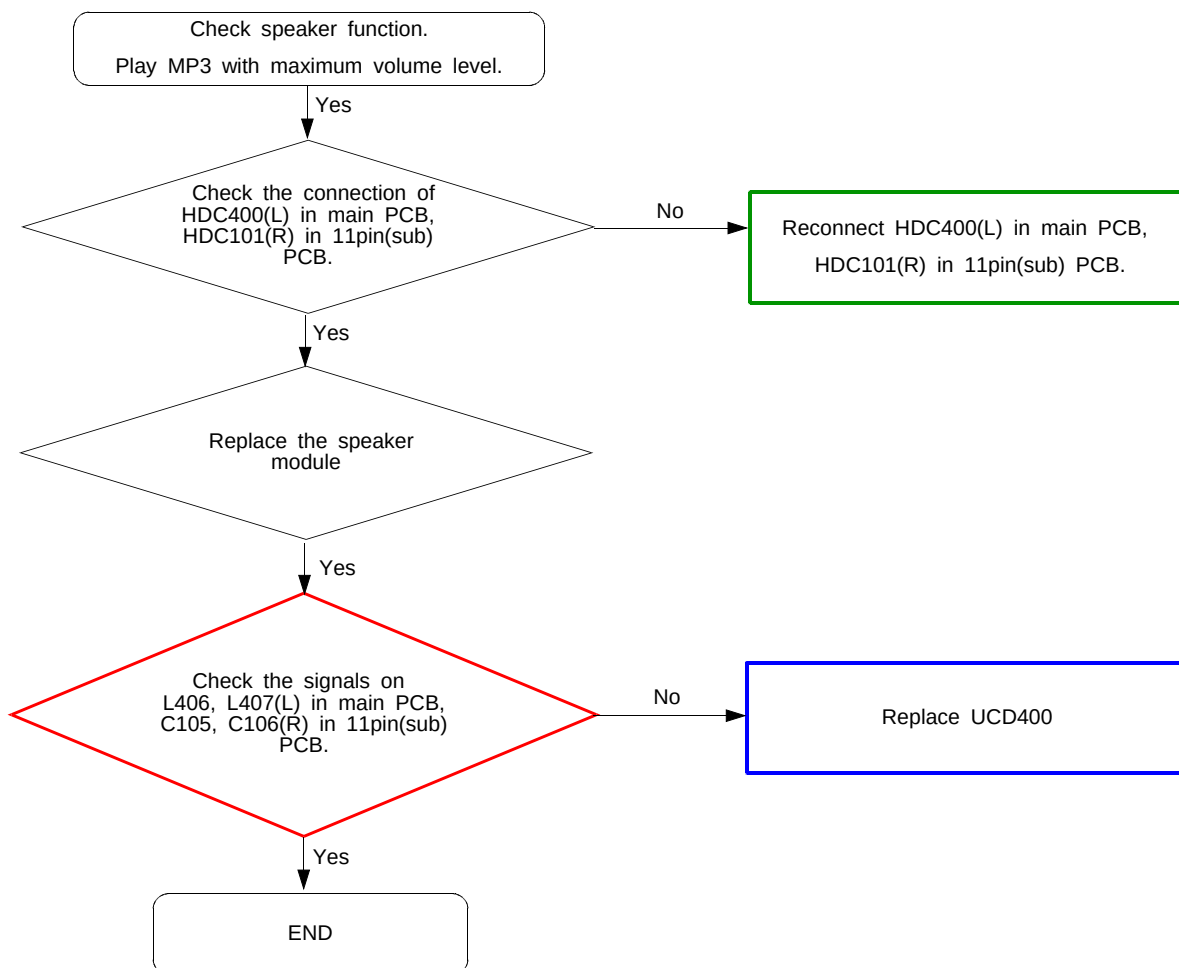


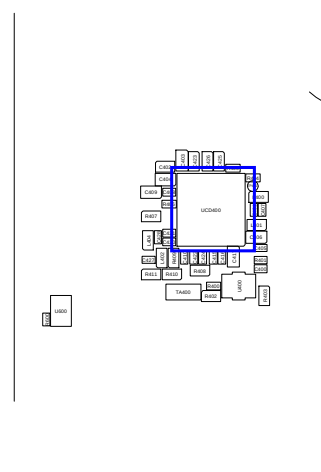
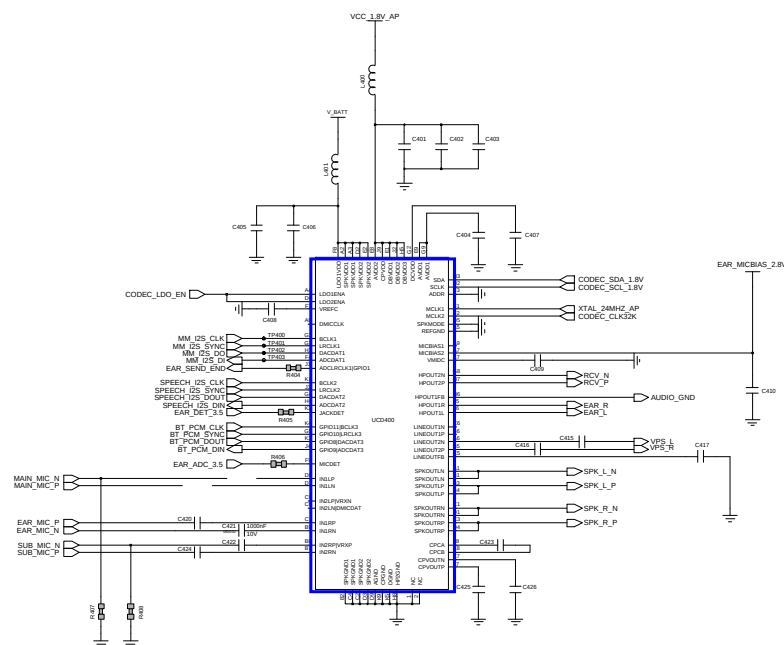
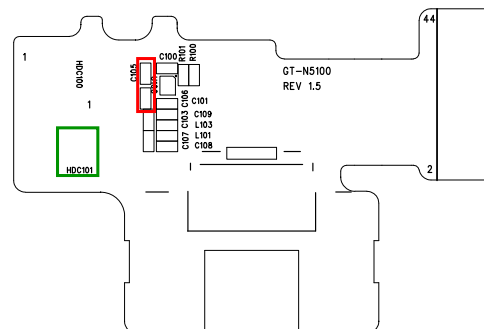
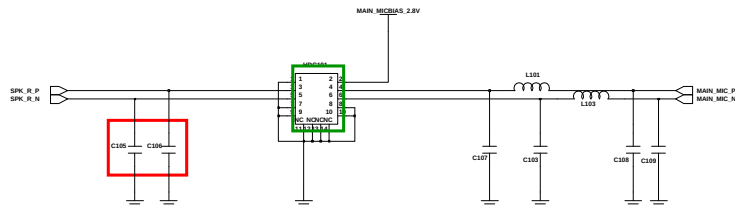
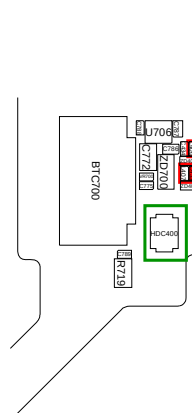
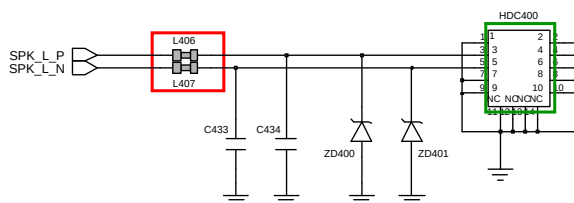
8-3-4. Microphone Part



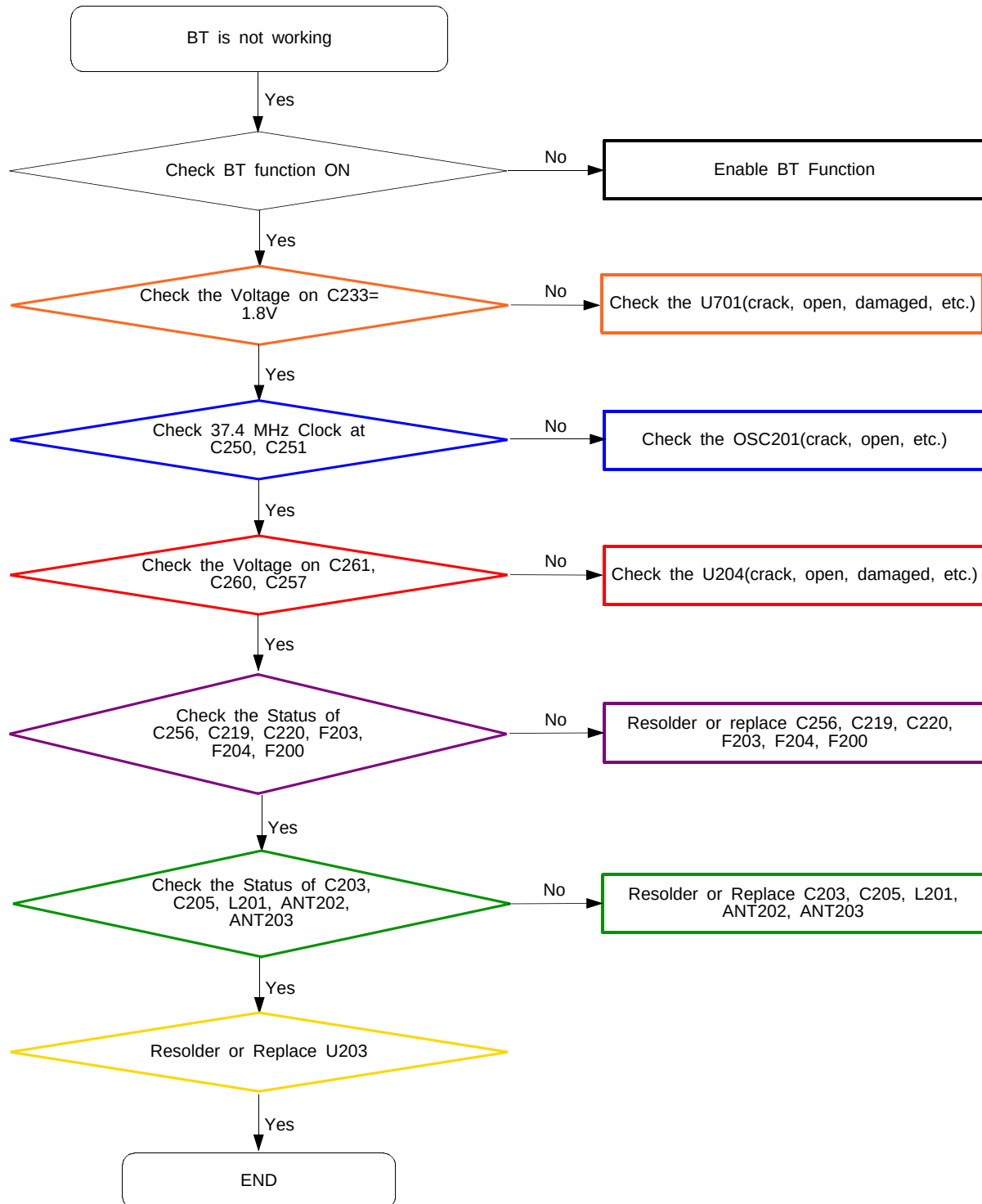


8-3-5. Speaker Part



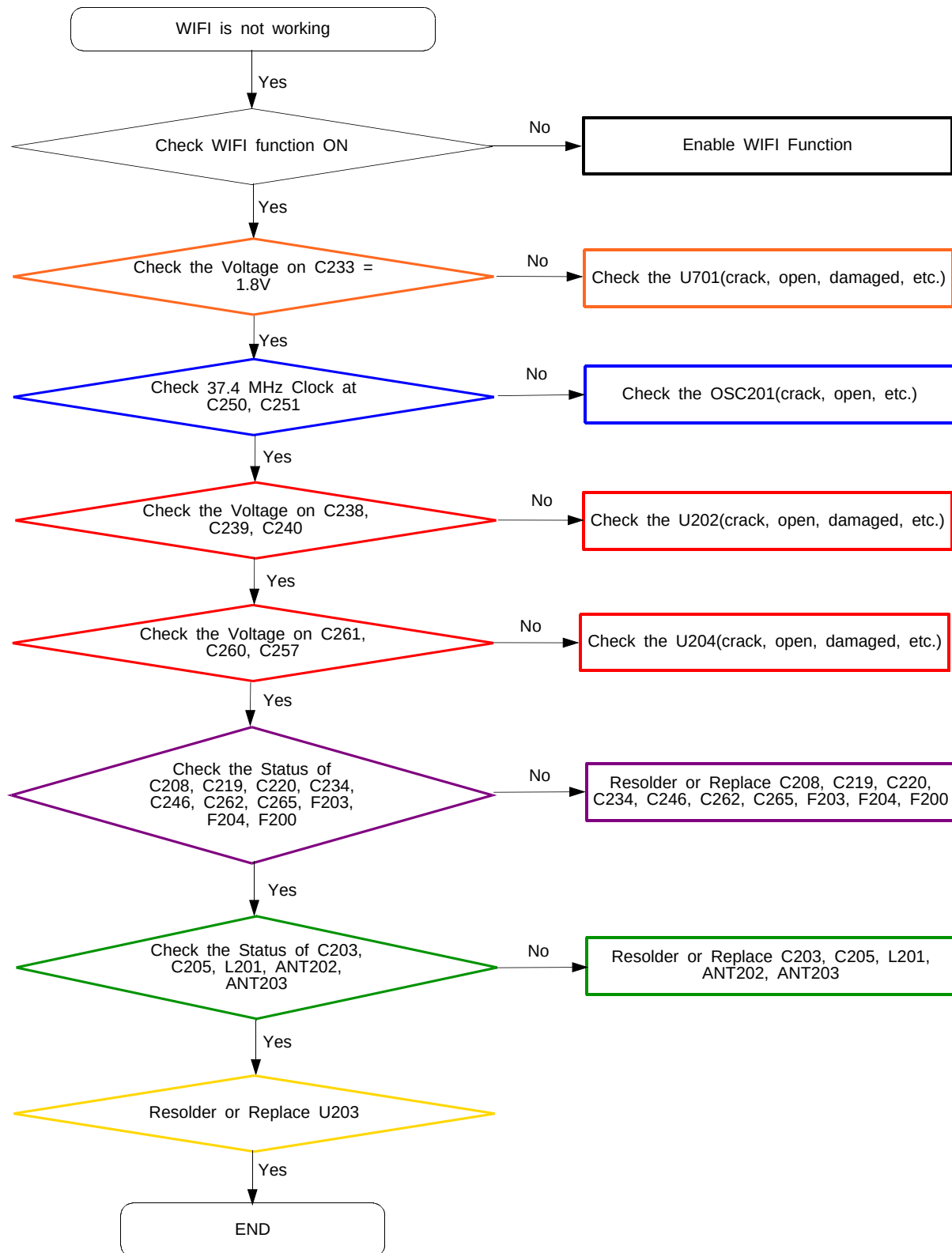


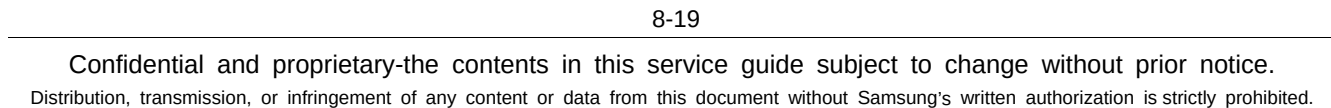
8-3-6. BT Part





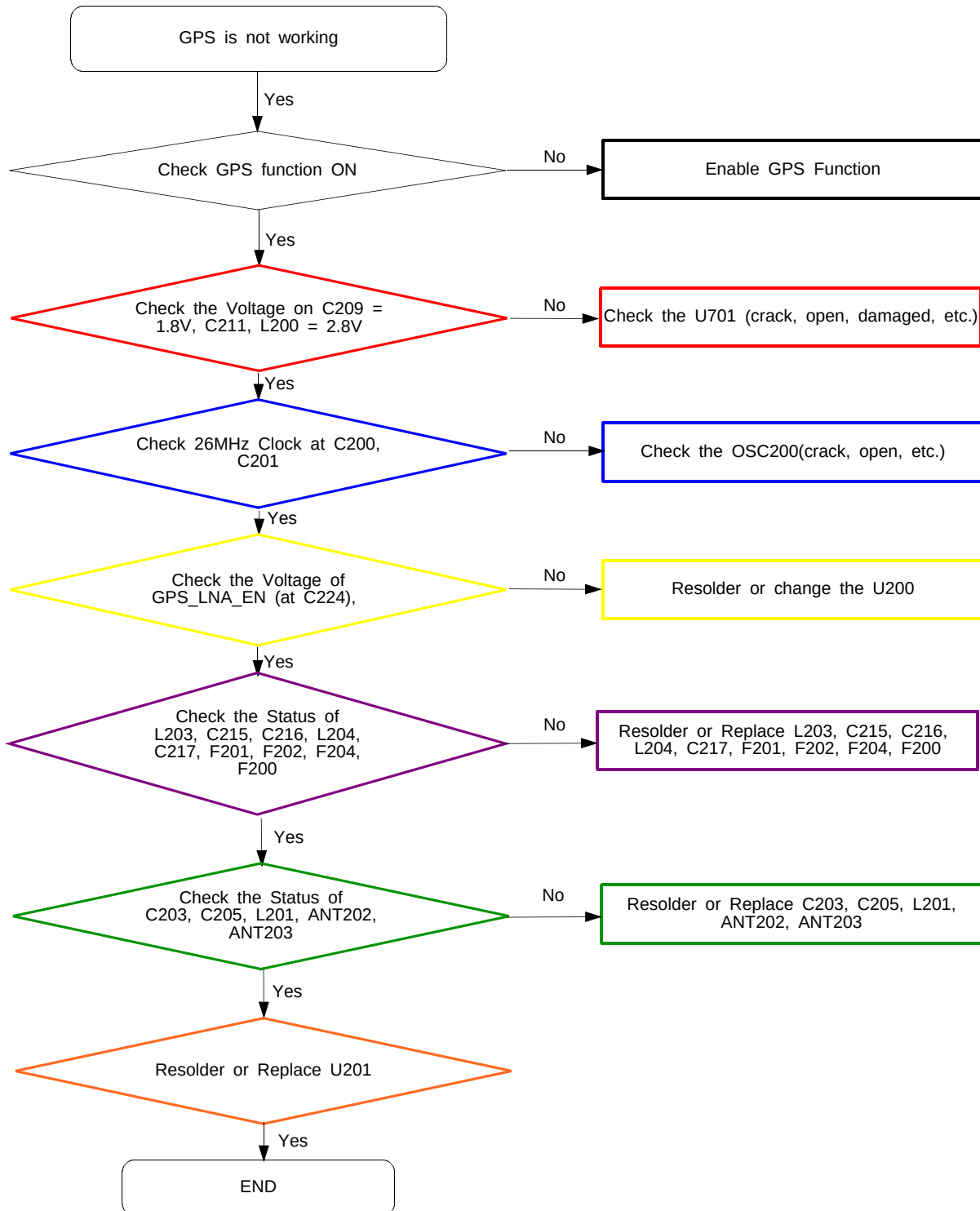
8-3-7. WIFI Part

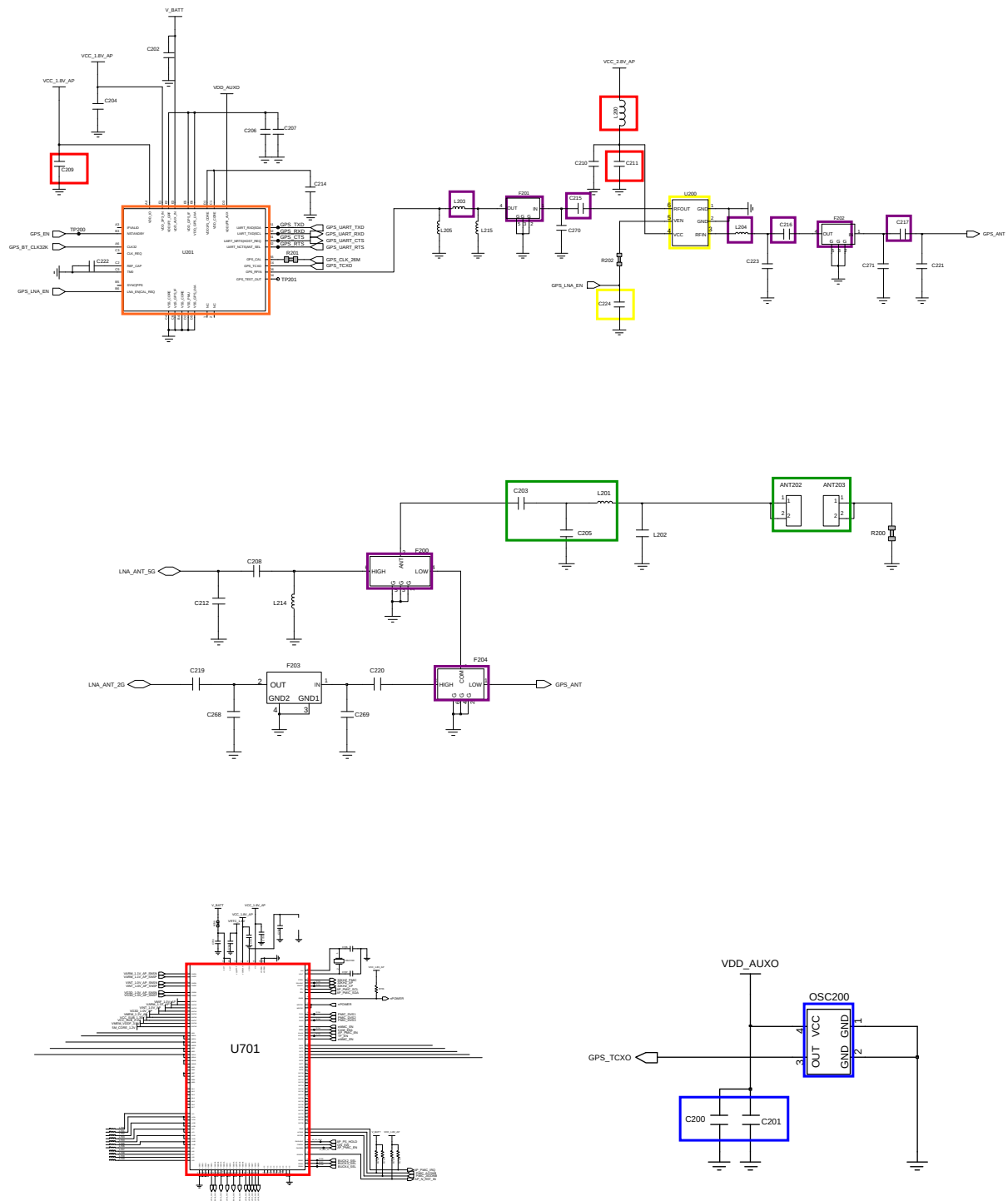


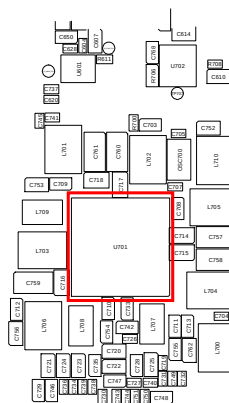
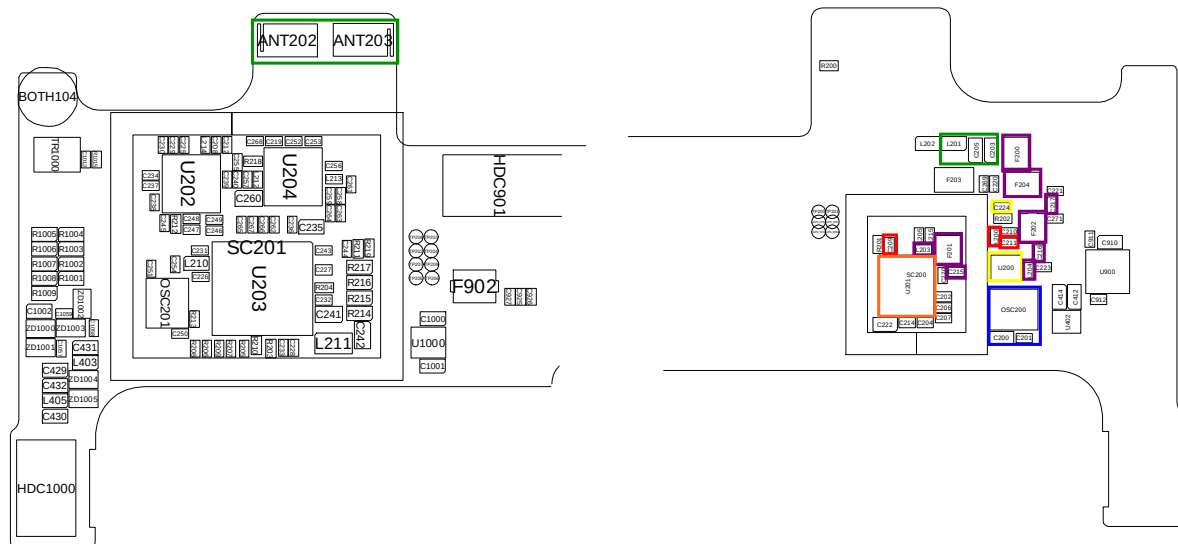




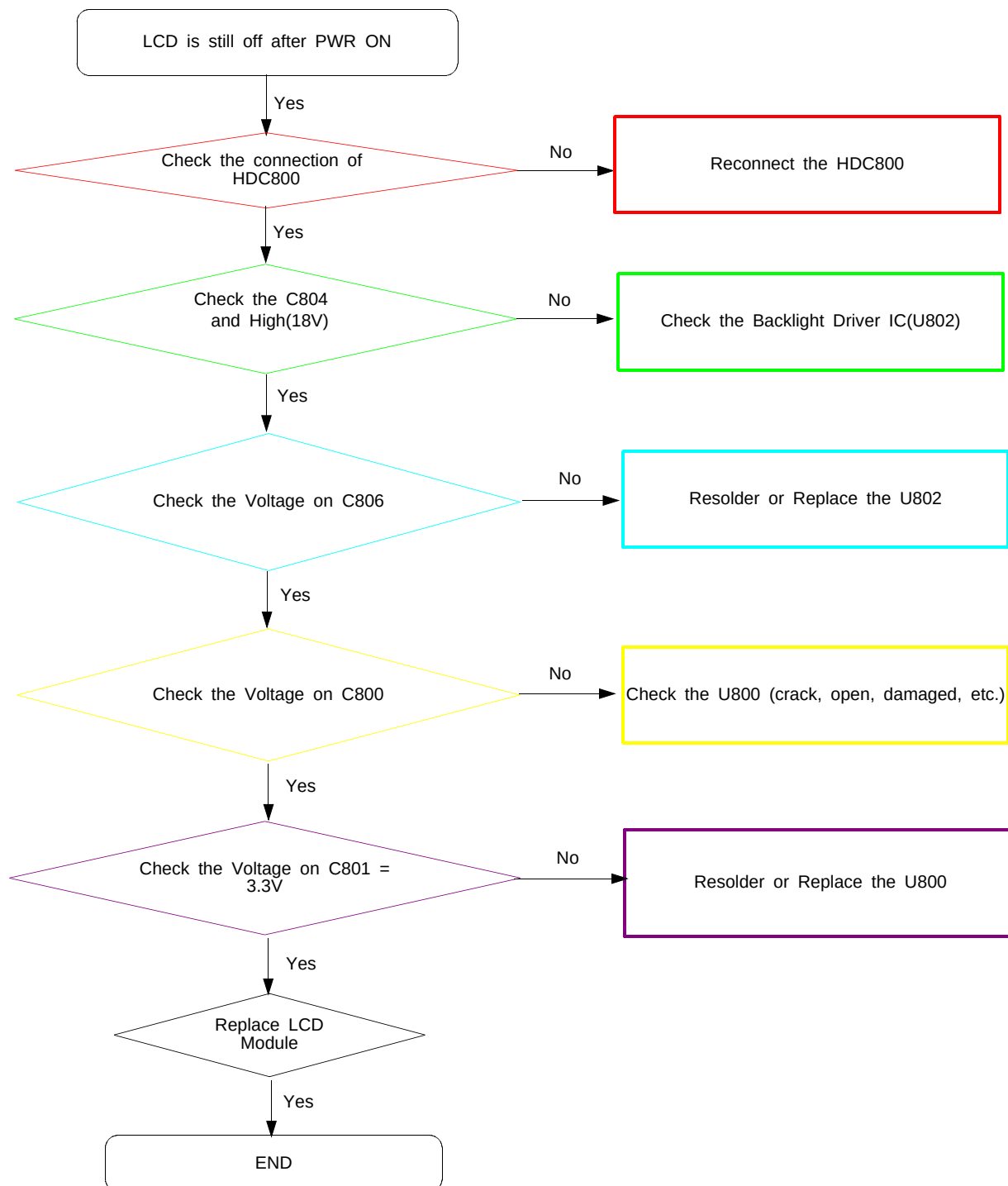
8-3-8. GPS Part

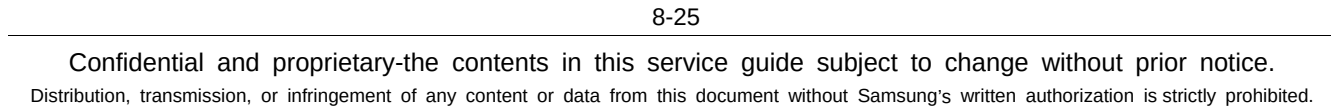




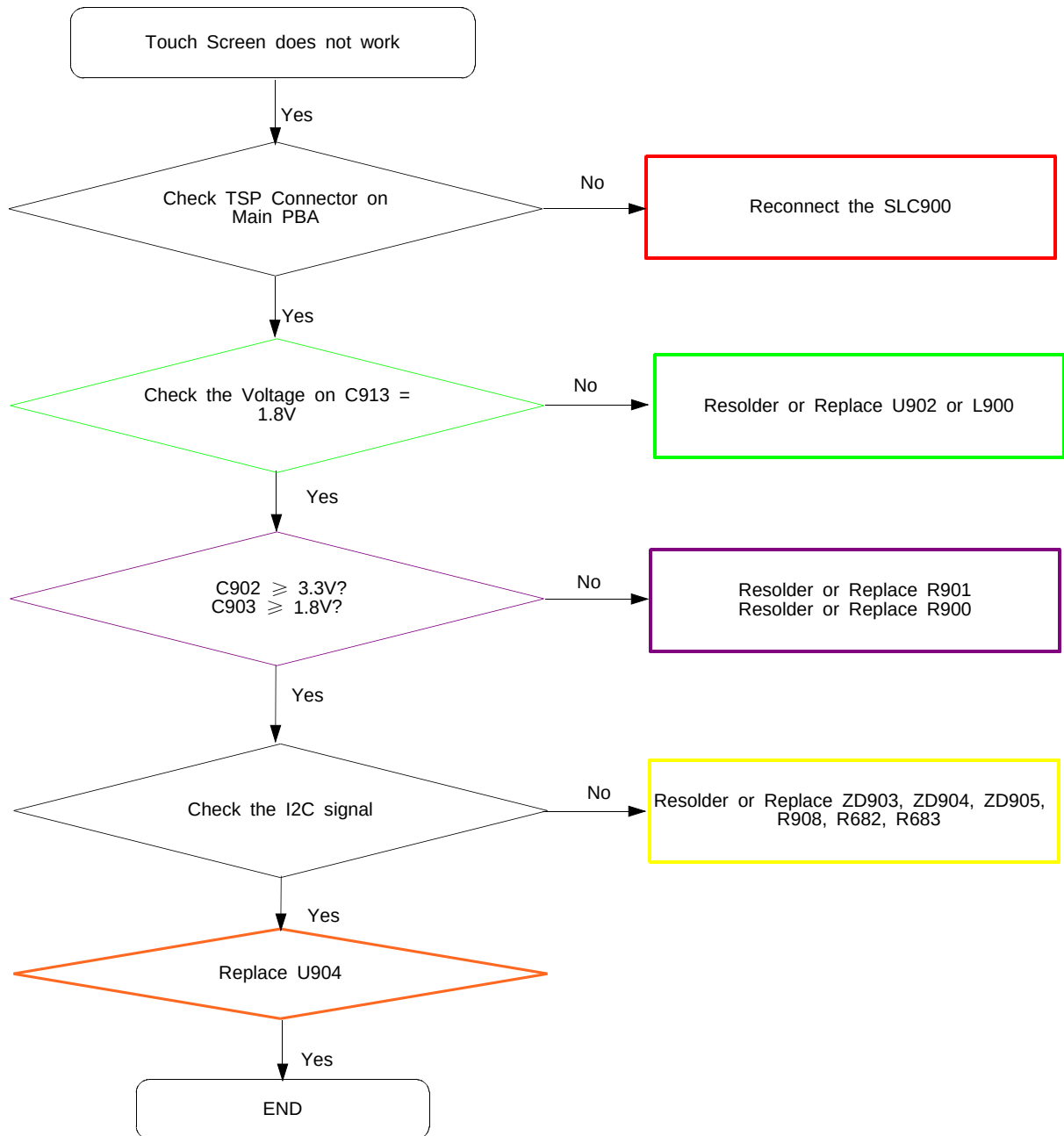


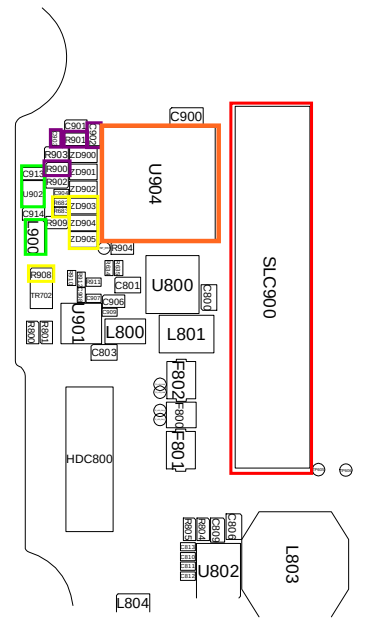
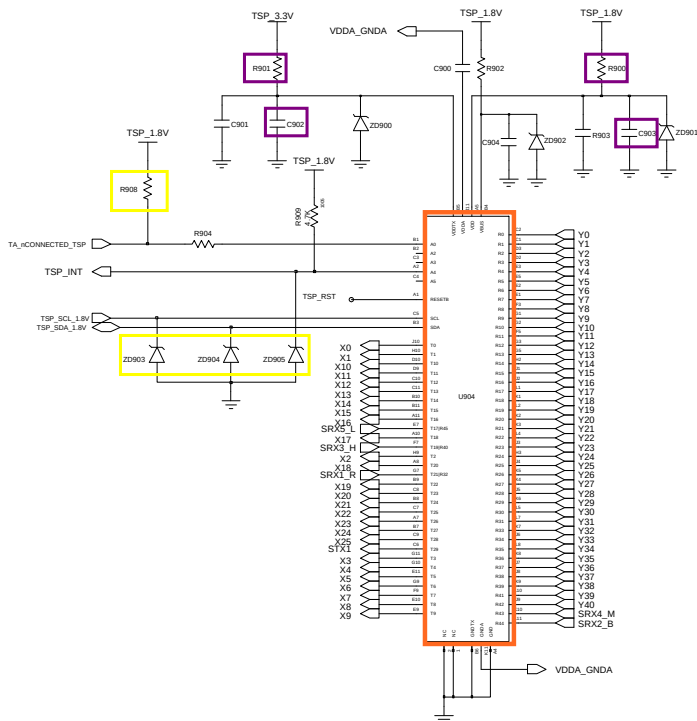
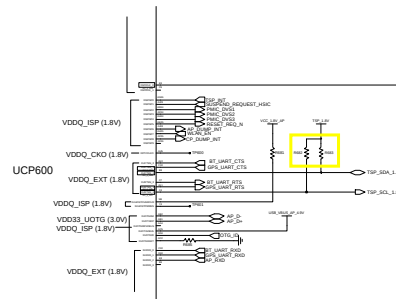
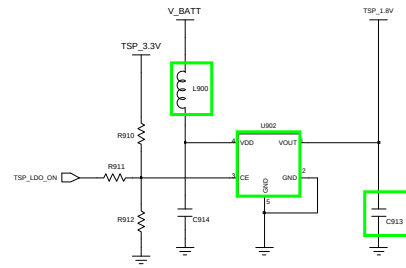
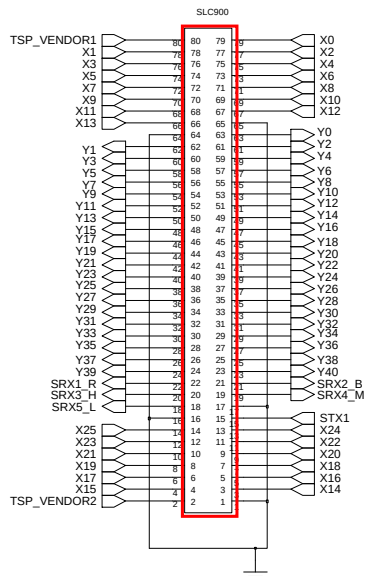
8-3-9. LCD



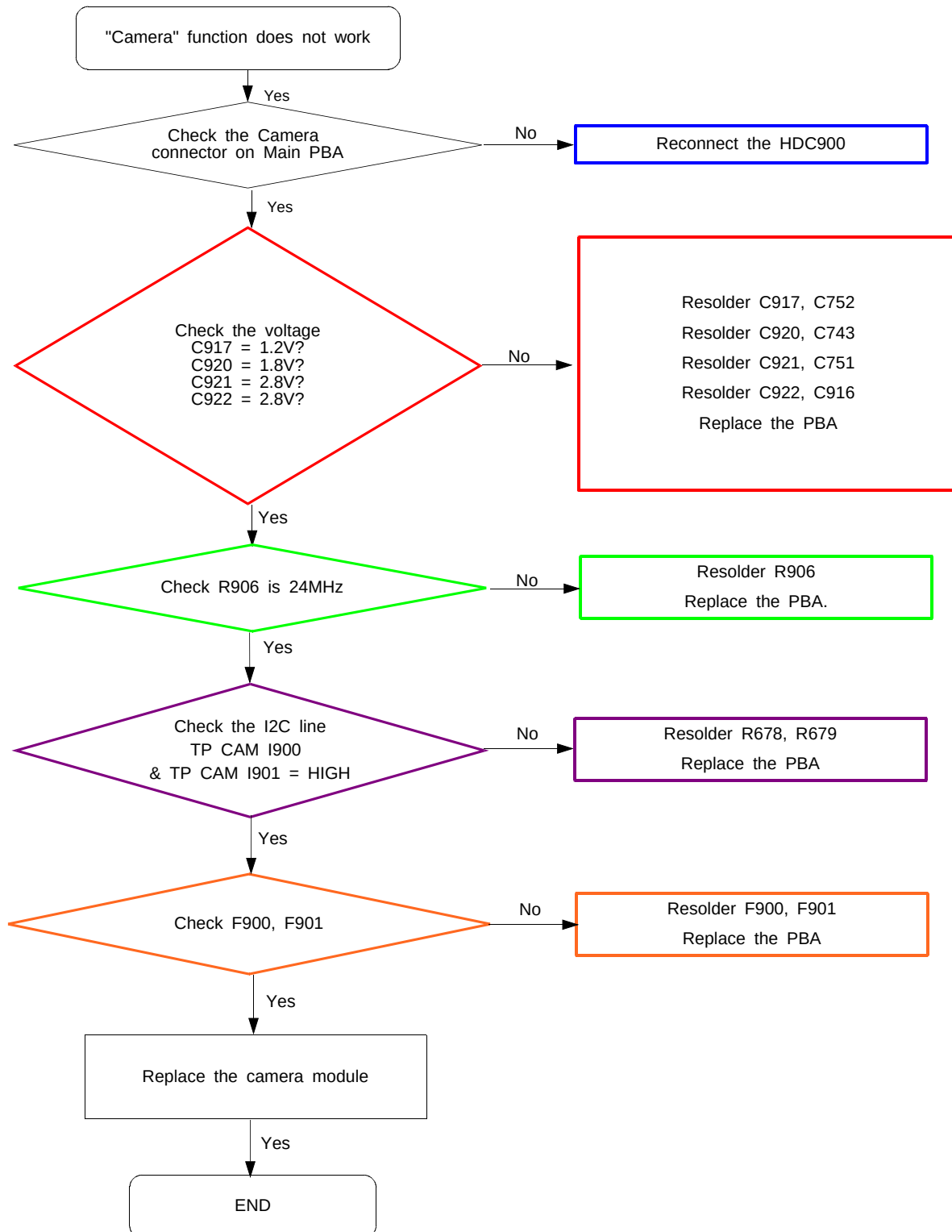


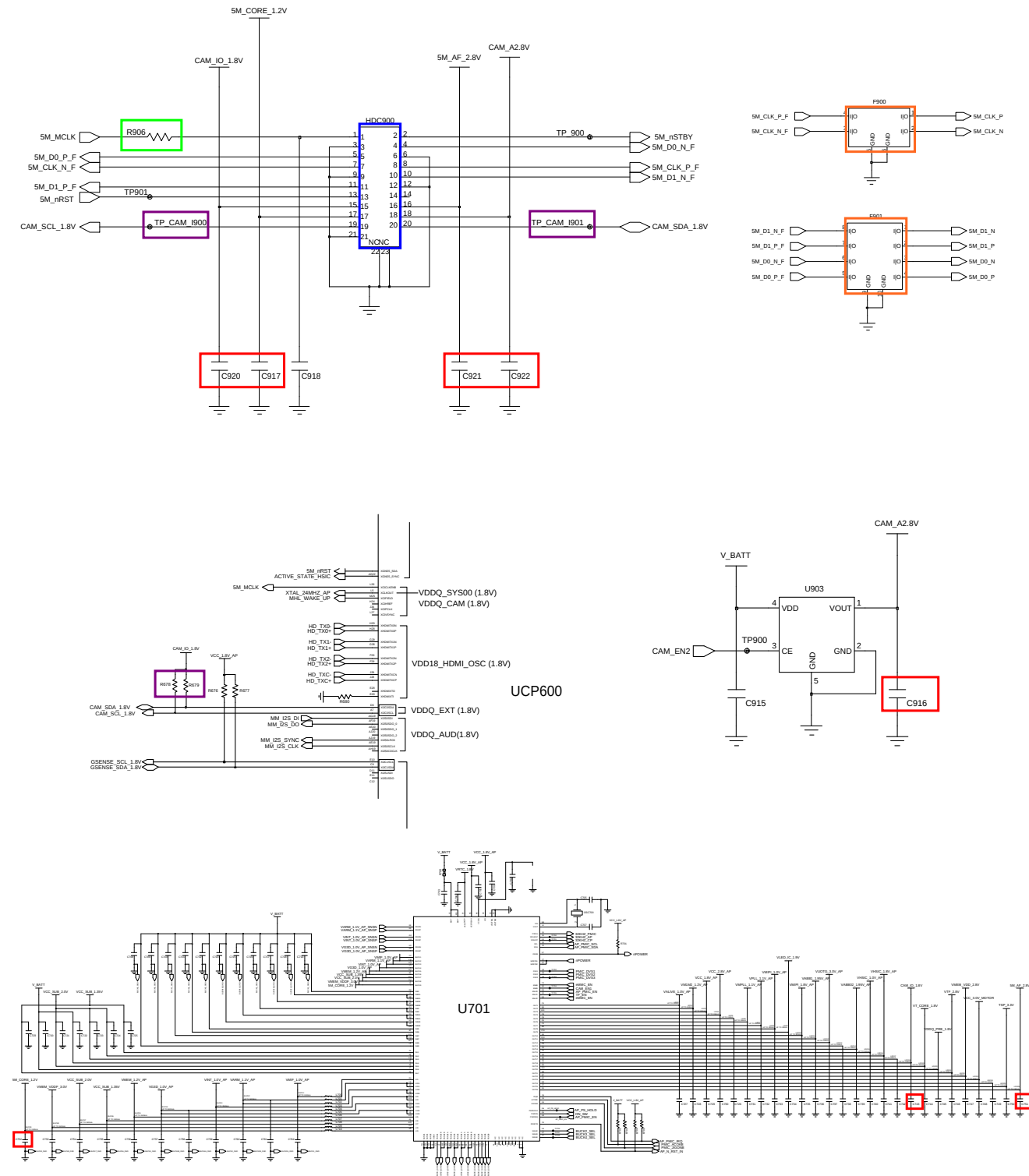
8-3-10. TSP

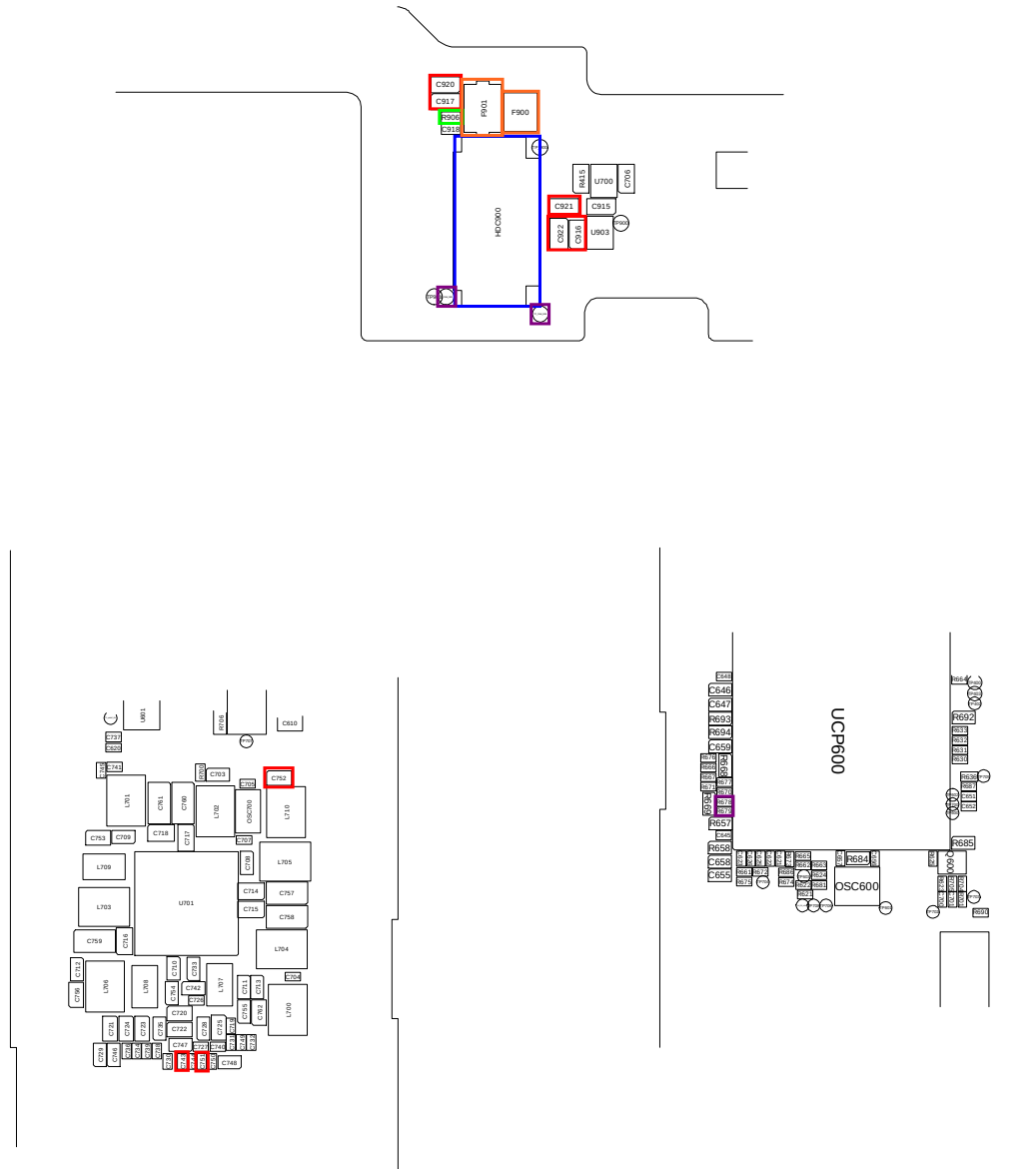




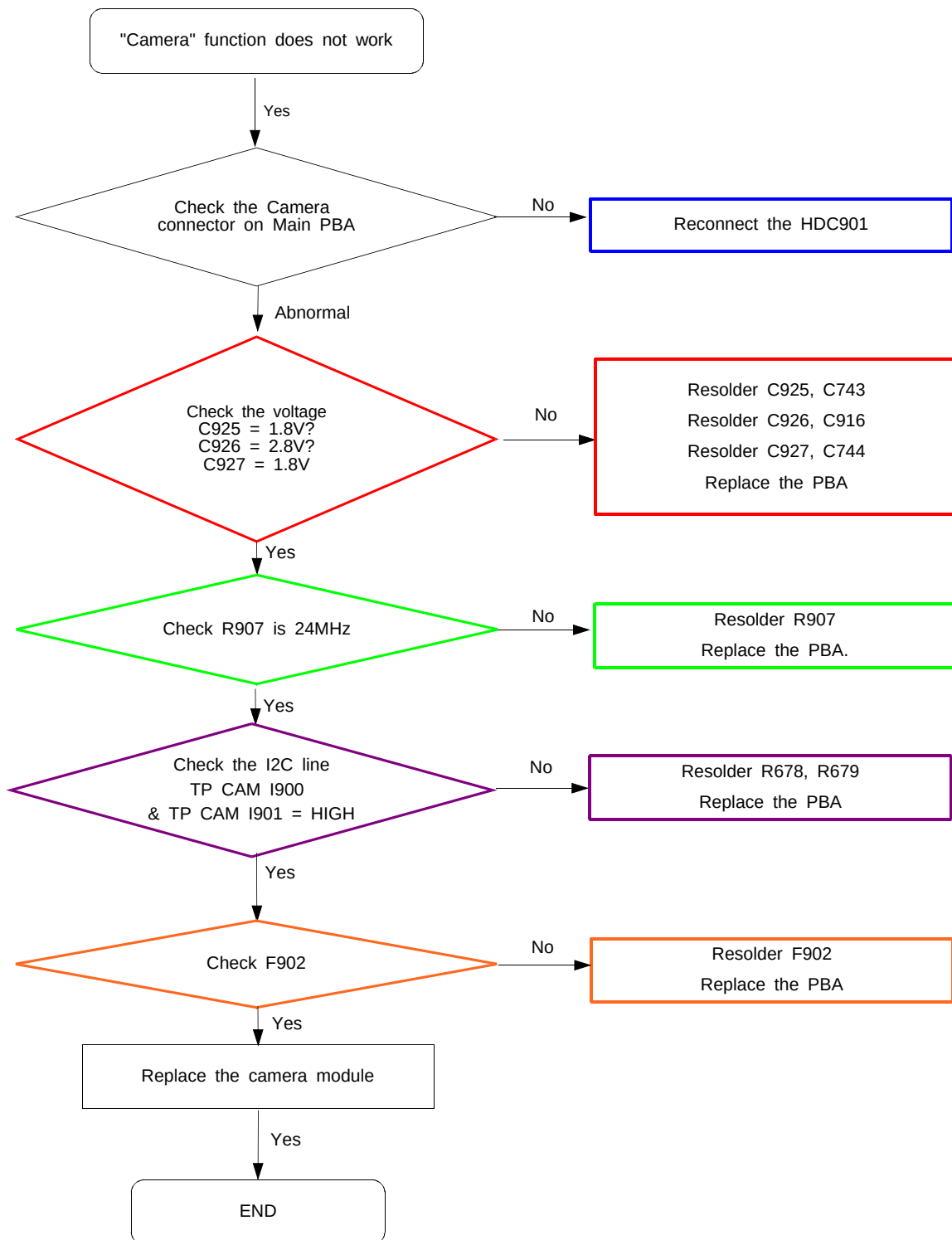
8-3-11. 3M CAM

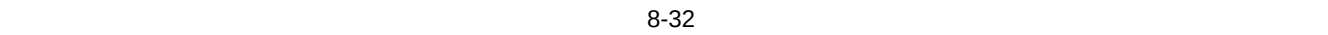
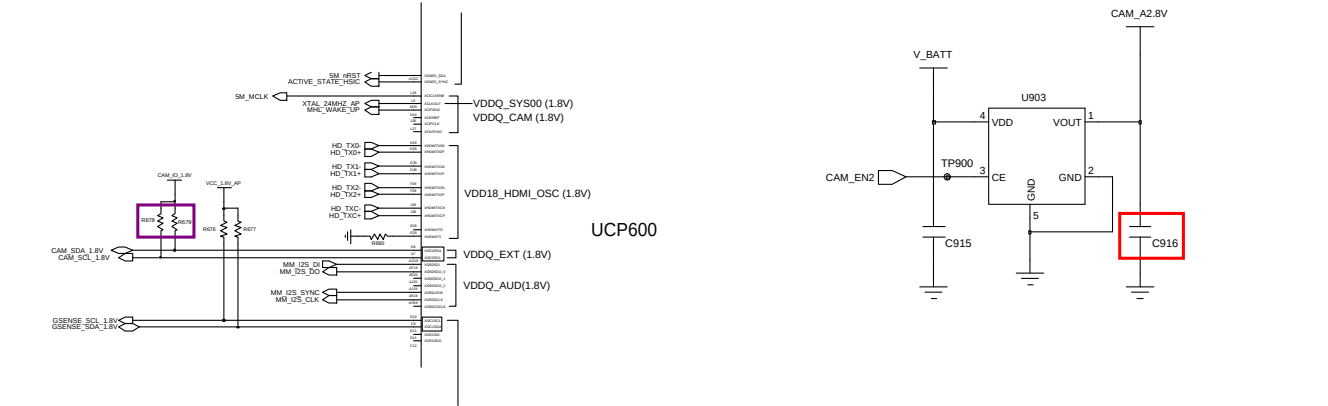


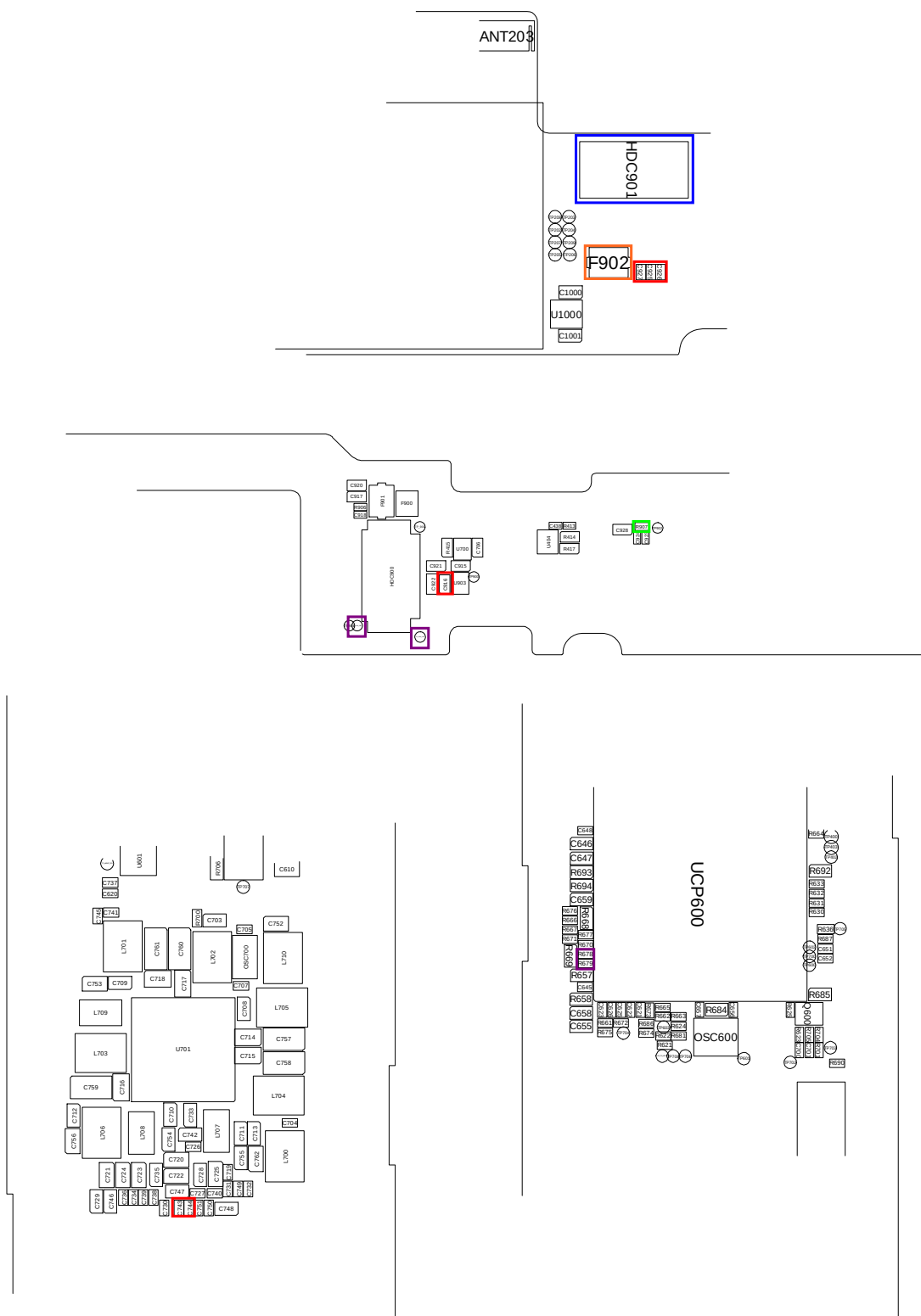




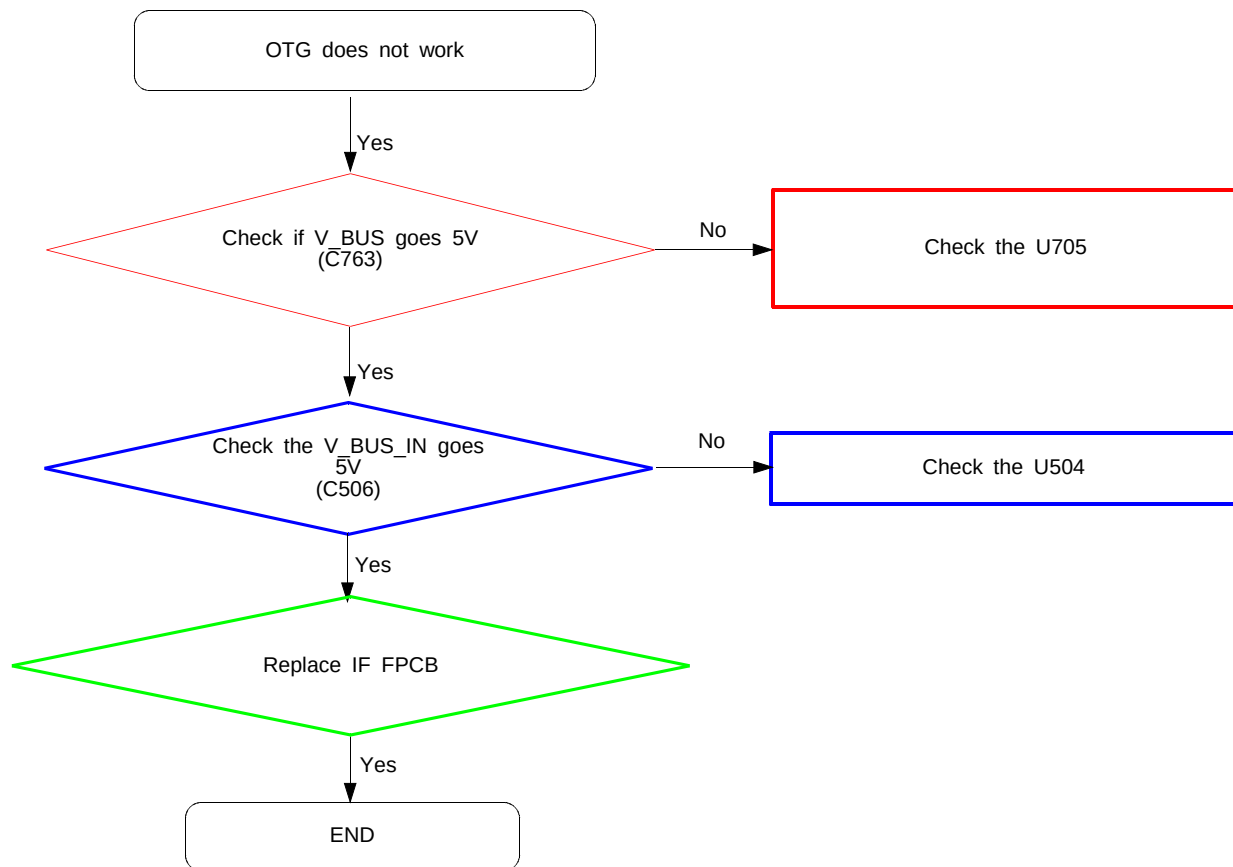
8-3-12. 1.3M CAM

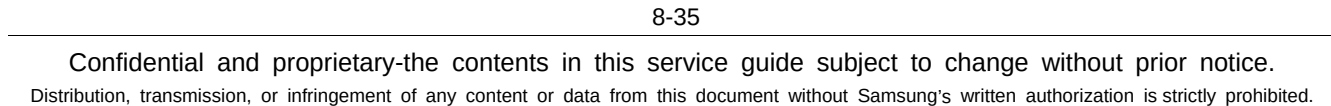




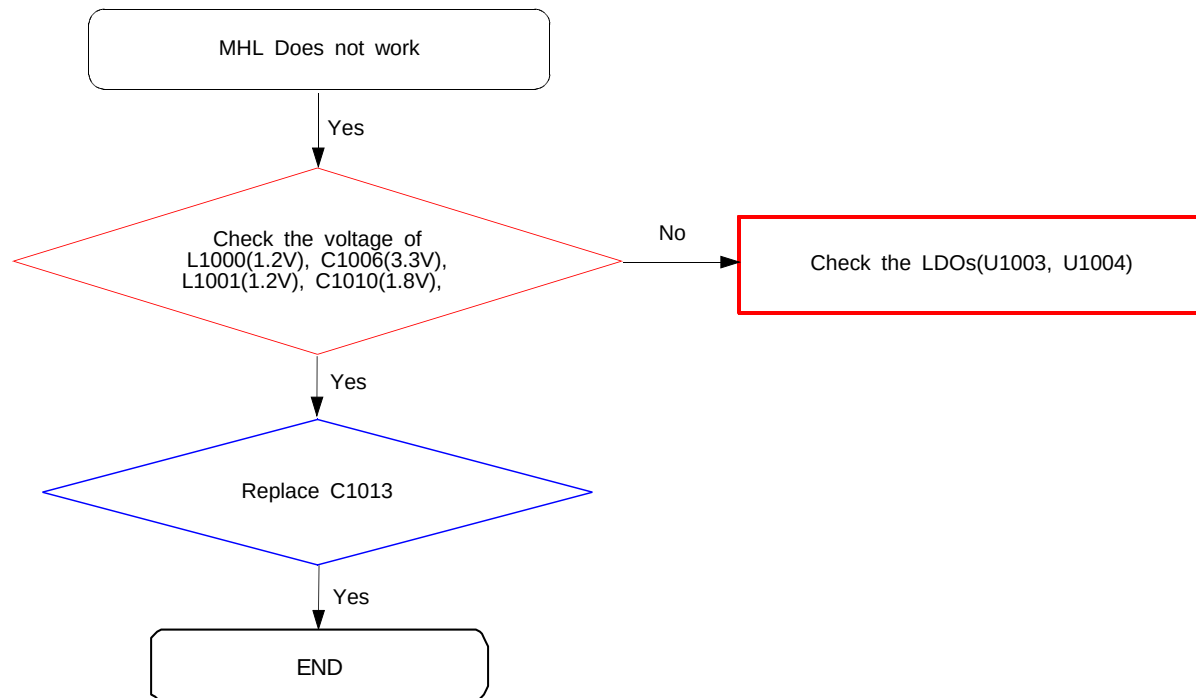


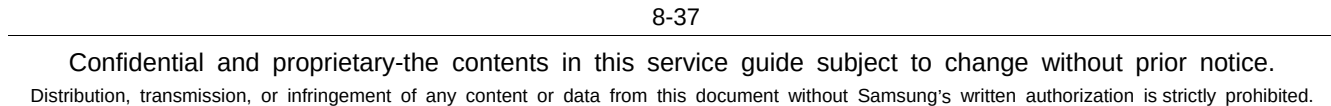
8-3-13. OTG





8-3-14. MHL





8-3-15. S-pen

